
Conformal Source Regions for Sparse-Sensor Pollution Localization: A Calibration, Exact-Posterior Audit, and Information Dose-Response Study

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Abstract

When a gas leak or pollution release happens, regulators and emergency responders must answer a deceptively simple question: *where is it coming from?* Increasingly this must be done from only a handful of cheap, fixed sensors. A machine-learning model can output a heatmap of likely source locations, but a heatmap alone does not say how much to trust it. We study whether a learned localizer can be equipped with *honest, trustworthy* uncertainty. We combine a standard permutation-invariant neural network (DeepSets) with split conformal prediction to produce spatial regions that provably contain the true source with a user-chosen probability (here 90%), and we *audit* those regions against the exact Bayesian posterior of a controlled two-dimensional advection–diffusion simulator. We preregistered three questions as hypotheses: are the regions *calibrated* (H1); are they as *sharp* as the data physically permit (H2, measured against the exact posterior); and do they respond correctly when we *add sensors, move sensors, or change noise* (H3, a controlled information “dose–response” test)? After the compute run we find: the conformal machinery is sound—applied to the exact posterior it attains 90.7% coverage, and the learned localizer attains 89.6% (87.5% for a second seed, a quantified finite-sample fluctuation); the learned regions strongly track the physics ordering (Spearman 0.85) but are a median $5.6\times$ larger than the exact-posterior regions—the network, not the data, is the sharpness bottleneck; and the learned regions respond in the correct direction on every information knob but with roughly half the oracle’s magnitude. The run also surfaced three implementation lessons that any similar pipeline must absorb, including a floating-point failure mode that silently destroys conformal coverage on sharp posteriors. To our knowledge this is the first study to pair finite-sample conformal spatial regions with an exact-posterior sharpness audit in this domain, and the first to test learned localization uncertainty causally through a controlled dose–response protocol.

1 Introduction

Locating the source of an airborne pollutant from a sparse network of fixed sensors is a classic inverse problem whose legal and public-health stakes are rising quickly. Under the U.S. EPA Super-Emitter Program, codified in New Source Performance Standards subpart OOOOb and Emissions Guidelines OOOOc (final rule at 89 FR 16820, published March 8, 2024), EPA-certified third parties notify operators of remotely detected methane emissions events, where a super-emitter event is defined as “any emissions event that is located at or near an oil and natural gas facility . . . that is detected using remote detection methods and has a quantified emission rate of 100 kg/hr of methane or greater” [U.S. EPA, 2024]. A recipient who owns or operates a facility within 50 meters of the reported latitude and

longitude must investigate and report the event under 40 CFR 60.5371b. (EPA subsequently issued deadline extensions delaying implementation of the Super-Emitter Program; interim final rule 90 FR 35966, July 31, 2025.) In the European Union, Regulation (EU) 2024/1787 (adopted 13 June 2024, in force 4 August 2024) mandates measurement, reporting, and verification of energy-sector methane, building on the higher reporting tiers of the Oil and Gas Methane Partnership 2.0 (OGMP 2.0) framework [European Union, 2024]. Both regimes support the Global Methane Pledge, described by UNEP as an attempt by 150 countries, led by the EU and US, to reduce global methane emissions by 30% by 2030 from 2020 levels. The UNEP International Methane Emissions Observatory’s Methane Alert and Response System (MARS) already issues satellite-based notifications globally: UNEP reported that MARS delivered 1,200 notifications to governments and companies over its first two years, of which just one per cent were responded to [UNEP, 2024].

In every one of these settings, attributing an emission to a *location* carries consequences, so the *uncertainty* of that attribution must itself be trustworthy. A localizer that is confidently wrong is worse than one that honestly reports “somewhere in this larger region.” Three properties matter. **Calibration** (trustworthiness): if the tool claims 90% confidence, the true source should fall inside its region 90% of the time. **Sharpness** (actionability): among calibrated regions, smaller is better, and a region should be as tight as the data physically allow — a large region is itself a signal that the sensor network is inadequate. **Dose–response stability** (trust under change): sensors are added and lost constantly, so a region should shrink when more information arrives and grow when it is removed, tracking what an oracle would do.

Contribution. We contribute a benchmark and an evaluation protocol, *not* a new algorithm. We deliberately assemble off-the-shelf components — DeepSets [Zaheer et al., 2017], split conformal prediction [Angelopoulos and Bates, 2021], and textbook grid-based Bayesian inversion — and ask whether the resulting uncertainty machinery is calibrated, sharp, and stable. We claim novelty only in the domain pairing, the audit protocol, the dose–response protocol, and the released artifact. Verbatim: *To our knowledge, this is the first study to combine finite-sample conformal spatial regions with an exact-posterior sharpness audit for sparse-sensor atmospheric advection–diffusion source localization, the first to test learned localization uncertainty causally via a controlled information dose–response protocol against exact posteriors, and the first public benchmark in this domain to ship exact per-scenario posteriors.* We claim no new conformal algorithm, no new architecture, no coverage under distribution shift, no field deployability, and no abstention capability.

2 Related Work

Source term estimation and Bayesian inversion. Estimating the location and rate of an atmospheric release from concentration measurements is reviewed comprehensively by Hutchinson et al. [2017]. Bayesian and Markov-chain Monte Carlo plume inversion [Newman et al., 2024] quantifies uncertainty rigorously but is per-scenario and computationally heavy, and its credibility depends on the correctness of the assumed dispersion model. Physics-informed neural networks recover advection–diffusion sources [Chuprov et al., 2025, Anague et al., 2025] but do not provide finite-sample coverage. The notion of “information density” in inverse problems [Bangerth et al., 2022] formalizes why some regions of a domain are recoverable and others are not; we return to it in our future-work ladder.

Conformal prediction for localization and inverse problems. Conformal prediction [Angelopoulos and Bates, 2021] yields distribution-free, finite-sample coverage and has recently reached localization: transductive conformal prediction for acoustic source localization with Gaussian processes [Rozenfeld and Laufer-Goldshtein, 2025]; task-driven conformal UQ for imaging inverse problems [Wen et al., 2024]; conformal multi-source detection on networks with recall guarantees [Jian et al., 2026]; and multi-output conformal regions for anatomical landmark localization [Jonkers et al., 2025]. Edwards and Smith [2025] use a categorical neural network (partitioning the domain into classes) and a Bayesian neural network for radiological plume source identification. Shi et al. [2026] (Distill-Belief) distill a Bayes-correct particle-filter teacher into a student for closed-loop inverse source localization. Most closely related in machinery, CP4SBI [Cabezas et al., 2025] applies local conformal calibration to simulation-based-inference credible sets, but on generic SBI benchmarks and a neuroscience model, with no atmospheric domain, no exact-posterior sharpness audit, and no dose–response protocol.

Table 1: Positioning matrix. $\checkmark$ = addressed; $(\checkmark)$ = partially/indirectly. Axes: A amortized inference, B finite-sample region coverage, C audit vs. exact/reference posterior, D controlled causal test of uncertainty, E sparse atmospheric sensors.

Work	A	B	C	D	E
PINN plume inversion [Chuprov et al., 2025, Anague et al., 2025]	$(\checkmark)$				$\checkmark$
Bayesian/MCMC STE [Newman et al., 2024]			$(\checkmark)$		$\checkmark$
Conformal acoustic loc. [Rozenfeld and Laufer-Goldshtein, 2025]	$(\checkmark)$	$\checkmark$			
Conformal imaging UQ [Wen et al., 2024]	$(\checkmark)$	$\checkmark$			
Conformal network detection [Jian et al., 2026]	$(\checkmark)$	$\checkmark$			
Distill-Belief [Shi et al., 2026]	$\checkmark$		$\checkmark$		$(\checkmark)$
CP4SBI [Cabezas et al., 2025]	$\checkmark$	$\checkmark$			
This work	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

Positioning. We organize the field along five axes: (A) fast amortized inference; (B) finite-sample region coverage; (C) audit against an exact/reference posterior; (D) controlled causal test of uncertainty; (E) sparse atmospheric sensors. PINN inversion occupies (E) and partially (A); MCMC plume inversion occupies (C-like) and (E); conformal acoustic/imaging/network localization occupy (A,B) in other domains; Distill-Belief occupies (A,C) for closed-loop sensing. Our study is the first to occupy (A)–(E) simultaneously in the sparse-sensor atmospheric advection–diffusion setting (Table 1).

3 Problem Setup and Physics

We consider a scalar pollutant concentration $C(\mathbf{x}, t)$ on the unit square $\Omega = [0, 1]^2$, $t \in (0, 1]$, transported by a constant wind $\mathbf{u} = (u, v)$ and spread by isotropic diffusion with coefficient D . The governing equation is the constant-coefficient advection–diffusion PDE with source term S :

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D \left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} \right) + S(\mathbf{x}, t), \quad \mathbf{x} \in \Omega, t \in (0, 1]. \quad (1)$$

For a continuous point release of unit rate at source location $\mathbf{x}_s$, switched on at $t = 0$, the field admits an analytic Green’s-function response obtained by superposing spreading Gaussian blobs advected downwind. Writing $\mathbf{u} = (u, v)$ and using a small source blob of width $\sigma_s = 1/128$ to regularize the singular point source, the unit-rate response is

$$A(\mathbf{x}, t; \mathbf{x}_s, \mathbf{u}, D) = \int_0^t \frac{1}{\pi w_s} \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_s - \mathbf{u} s\|^2}{w_s}\right) ds, \quad w_s = 4Ds + 2\sigma_s^2. \quad (2)$$

The age integral in Eq. (2) is evaluated by 64-node Gauss–Legendre quadrature after a log-age substitution $s = t e^{-\tau}$, $\tau \in [0, 12]$, that concentrates nodes near the near-singular small- s region (the truncation bound $\tau_{\max} = 12$ was tuned empirically; see Appendix A); responses are generated in `float32` and validated in `float64` (Appendix A). A source of physical rate q produces field qA .

Observation model. Sensor i at position $\mathbf{x}_i$ takes 30 readings at times $t_k = k/30$, $k = 1, \dots, 30$. Each reading is a noisy, reference-normalized concentration

$$y_{ik} = \frac{q A(\mathbf{x}_i, t_k; \mathbf{x}_s, \mathbf{u}, D)}{C_{\text{ref}}} + \varepsilon_{ik}, \quad \varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2), \quad (3)$$

with independent Bernoulli reading dropout: a dropped reading is simply *omitted* (never imputed), so the model must handle variable-length, irregularly-sampled inputs. The complete data-generation prior is given in Table 2. Splits are always by *complete scenario* — readings from a single scenario are never divided across train/validation/calibration/test — to prevent leakage (Section 5).

4 Method

4.1 Amortized posterior (DeepSets classifier)

The localizer is a permutation-invariant DeepSets network [Zaheer et al., 2017] of roughly 5M parameters (4,869,568 as instantiated), chosen deliberately as a standard, unremarkable architecture.

Table 2: Benchmark data-generation prior. All quantities are sampled independently per scenario. $\text{LogUniform}(a, b)$ is uniform in log-space.

Quantity	Symbol	Distribution
Source location	$\mathbf{x}_s$	$\text{Uniform}([0.1, 0.9]^2)$
Release rate	q	$\text{LogUniform}(0.5, 5)$
Wind speed	$\ \mathbf{u}\ $	$\text{Uniform}(0.5, 2.0)$
Wind direction	θ_u	$\text{Uniform}(0, 2\pi)$
Diffusivity	D	$\text{LogUniform}(0.002, 0.02)$
Sensor count	N	Uniform integer $\{4, \dots, 12\}$
Sensor positions	$\mathbf{x}_i$	i.i.d. $\text{Uniform}([0, 1]^2)$
Noise std	σ	$\text{LogUniform}(0.01, 0.10)$
Reading dropout	p_{drop}	$\text{Uniform}(0, 0.20)$
Observation times	t_k	$k/30, k = 1, \dots, 30$
Train / Val / Calib / Test		10,000 / 1,000 / 2,000 / 2,000
Dose-response base scenarios		50 (12-sensor master layouts)

Each retained reading is tokenized as

$$\mathbf{z}_{ik} = [\text{Fourier}(x_i), \text{Fourier}(y_i), \text{Fourier}(t_k), \text{asinh}(y_{ik}/\sigma)], \quad (4)$$

with frequency set $\{1, 2, 4, 8\}$ for the Fourier features and the asinh transform compressing the heavy-tailed concentration scale. A per-token MLP (input $\rightarrow 256 \rightarrow 256$, GELU + LayerNorm) embeds each token; embeddings are pooled by *both* masked mean and masked max and concatenated. A global context MLP ($5 \rightarrow 64 \rightarrow 64$) encodes $(u, v, \log D, \log \sigma, N)$. The decoder MLP ($576 \rightarrow 1024 \rightarrow 4096$ logits) produces a softmax $P_j(\cdot)$ over the 4096 cells of a 64×64 grid.

Training. We minimize cross-entropy to a smoothed target: a normalized Gaussian bump (std 0.75 cells, truncated at 3σ) on the grid centered at the true source. Optimization uses AdamW, learning rate 3×10^{-4} cosine-decayed to 3×10^{-5} , weight decay 10^{-4} , batch size 256 scenarios, up to 200 epochs, early stopping on validation negative log-likelihood (patience 25), gradient clipping at 1.0, mixed precision, two random seeds, checkpointing by validation NLL only. Because targets are sampled from the prior over sources, cross-entropy training makes the softmax an *amortized approximation to the grid posterior* $p(c | \mathbf{y})$, subject to the usual caveats of finite capacity, finite data, and grid discretization.

Amendment: symmetry augmentation (deviation from the preregistered recipe). As preregistered, the recipe above *memorizes* the 10,000 training scenarios with zero generalization: training NLL falls to 1.5 while validation NLL never improves below the near-uniform level (≈ 7.96 ; $\log 4096 = 8.32$), and validation MAP error (0.44) is worse than the peak-sensor baseline (0.37) and no better than guessing from the prior (0.42). We verified this is not caused by mixed precision, learning rate, or the context branch. We therefore apply to each training batch a random D4 symmetry of the unit square (rotation by $k \cdot 90^\circ$ plus reflection) jointly to sensors, wind, and source. This is an *exact* physics symmetry (validated to 2×10^{-13} ; Appendix A) and the benchmark prior is D4-invariant, so augmented scenarios are exact draws from the same prior; data, architecture, loss, and all other hyperparameters are unchanged. With augmentation, validation NLL tracks training loss with no overfitting gap. We flag this as a finding: at this data scale the benchmark *requires* symmetry augmentation to be learnable by the specified architecture.

4.2 Conformal regions

Let P_j be the predicted distribution for calibration scenario j and c_j^* the true cell. We use a randomized highest-density-mass split conformal nonconformity score

$$s_j = \sum_{c: P_j(c) > P_j(c_j^*)} P_j(c) + U_j \cdot \sum_{c: P_j(c) = P_j(c_j^*)} P_j(c), \quad U_j \sim \text{Uniform}(0, 1), \quad (5)$$

i.e. the predicted mass strictly above the true cell plus a uniformly randomized fraction of the tied mass (randomization removes discreteness so exact coverage is attainable). With $n = 2000$ calibration

scenarios and miscoverage $\alpha = 0.10$, the threshold $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score. The conformal region for a test input $\mathbf{y}$ is the smallest set of top-ranked cells whose cumulative predicted mass reaches $\hat{q}$. Under exchangeability of calibration and test scenarios this region satisfies the marginal coverage guarantee

$$\Pr(c^* \in \mathcal{C}_\alpha(\mathbf{y})) \geq 1 - \alpha. \quad (6)$$

We state plainly what Eq. (6) is and is not: it is *marginal* (averaged over the test distribution), *in-distribution* (calibration and test drawn from the same prior), *not conditional* (no per-covariate guarantee), and it does *not* survive distribution shift.

Amendment: numerically safe implementation of Eq. (5). For sharp heatmaps the exceedance mass s_j sits at $1 - \epsilon$ with ϵ as small as 10^{-30} ; in floating point $1 - \epsilon$ rounds to exactly 1.0 (catastrophic cancellation), creating an atom of tied scores whose calibration quantile silently breaks the region rule: conformal-on-exact-posterior covered only 0.84 instead of 0.90 when Eq. (5) was evaluated naively (and 0.82 with `float32` posterior storage). We therefore compute throughout with the complement “tail mass” $t_j = \sum_{c: P_j(c) < P_j(c_j^*)} P_j(c) + (1 - U_j) \cdot (\text{tied mass})$, for which $s_j = 1 - t_j$ holds *exactly* in exact arithmetic and which is representable at both ends (t -space thresholds reach 4×10^{-199} on the exact posterior); regions are built by excluding the largest ascending-mass tail, summing small numbers first. All coverage semantics are unchanged. We flag this as a general caution for conformal prediction over sharp probability maps.

4.3 Exact-posterior oracle

For each candidate cell c with unit-rate response vector $\mathbf{g}_c = (A(\mathbf{x}_i, t_k; c, \mathbf{u}, D))_{ik}$ evaluated at the (known) meteorology, the Gaussian likelihood for rate q is

$$p(\mathbf{y} | c, q) = \mathcal{N}(\mathbf{y}; q \mathbf{g}_c, \sigma^2 \mathbf{I}). \quad (7)$$

We marginalize the rate over a `LogUniform(0.5, 5)` prior,

$$p(\mathbf{y} | c) = \int p(\mathbf{y} | c, q) \pi(q) dq, \quad p(c | \mathbf{y}) = \frac{p(\mathbf{y} | c)}{\sum_{c'} p(\mathbf{y} | c')}, \quad (8)$$

normalized over all 4096 cells of the 64×64 grid. This exact posterior is the reference against which learned regions are audited. It is an *oracle*: it uses the true meteorology and noise level, so it represents the tightest honest region the data permit.

Amendment: peak-aware rate quadrature (deviation from the preregistered rule). The preregistered 32-node Gauss–Legendre rule in $\log q$ is mathematically inadequate here: because $\|\mathbf{y} - q \mathbf{g}_c\|^2$ is quadratic in q , the integrand is a Gaussian in q of width $\sigma / \|\mathbf{g}_c\|$ centered at $\hat{q}_c = \langle \mathbf{y}, \mathbf{g}_c \rangle / \|\mathbf{g}_c\|^2$, up to $\sim 1000\times$ narrower than the fixed rule’s node spacing, producing log-posterior errors of many nats (the pipeline’s own brute-force gate fails). We instead split $[0.5, 5]$ per cell at $\hat{q}_c \pm 8 \sigma / \|\mathbf{g}_c\|$ (clipped) and place 32 Gauss–Legendre nodes on the peak panel and 16 on each outer panel (64 total), evaluated entirely in `float64` log-space via the expansion $\|\mathbf{y} - q \mathbf{g}_c\|^2 = \|\mathbf{y}\|^2 - 2q \langle \mathbf{y}, \mathbf{g}_c \rangle + q^2 \|\mathbf{g}_c\|^2$. The composite rule agrees with a $10\times$ -node deep rule and a 200,000-node trapezoid to $< 10^{-6}$ in log-posterior (Appendix B).

4.4 Dose–response protocol

To test uncertainty *causally* we manipulate information content on constructed scenario sets and measure how learned and exact region areas respond. **Knob 1 (sensor count):** nested subsets $N = 4, \dots, 12$ of a fixed 12-sensor master layout, added one at a time in a fixed order, so each added sensor is a controlled intervention. **Knob 2 (geometry):** matched pairs at $N = 6$ contrasting a stratified, well-spread layout against a confined single-corner-quadrant layout that induces directional ambiguity. **Knob 3 (noise):** a sweep of σ across $[0.01, 0.10]$ with regenerated readings. All manipulations stay within the training ranges (in-support but atypical). Dose–response sets use $p_{\text{drop}} = 0$ (the in-support boundary) so that knob interventions are not confounded by random reading loss.

Wording discipline. Constructed layouts are *not* exchangeable draws from the benchmark prior, so we report region-size behavior *only* on these sets. The coverage claim of Eq. (6) lives exclusively on the random in-distribution test split.

Table 3: Headline results. Coverage target $1 - \alpha = 0.90$; test split, $n = 2000$; seed 1 (seed 2 in Appendix C). Area as fraction of domain.

Metric	Learned + conformal	Exact posterior (oracle)	Peak-sensor
MAP error (mean)	0.174	0.160	0.356
MAP error (median)	0.103	0.023	0.315
Coverage @ 90%	0.875	0.907	—
Clopper–Pearson 95% CI	[0.860, 0.889]	[0.893, 0.919]	—
Region area (mean)	0.168	0.221	—
Region area (median)	0.068	0.013	—

5 Experimental Protocol and Preregistered Analyses

Baselines. (i) A *peak-sensor heuristic* that places a point estimate at the sensor with the largest time-integrated concentration (estimated as the mean of retained readings, fair across dropout rates). (ii) The *exact posterior* (oracle).

Preregistered hypotheses. **H1 (calibration).** Conformal regions attain verified 90% marginal coverage on the in-distribution test split, reported with an exact Clopper–Pearson 95% interval. **H2 (sharpness).** Learned region area tracks exact-posterior area at matched coverage. We apply the *identical* conformal procedure to the exact posteriors so both learned and exact regions sit at the same verified 90% coverage, then report the per-scenario Spearman correlation of areas and the mean/median area ratio (learned/exact), which we call the *inefficiency factor*. **H3 (dose–response, causal).** Under each knob, learned region area responds monotonically in the same direction and comparable proportion as exact-posterior area; we report contraction-slope ratios and per-knob Spearman agreement. **D1 (descriptive).** A scatter of learned region area versus MAP localization error, with one sentence on future abstention.

Five headline metrics. (1) MAP localization error (mean, median); (2) empirical coverage at 90% with Clopper–Pearson interval; (3) mean/median region area as a fraction of the domain; (4) sharpness agreement (H2 Spearman + inefficiency factor); (5) dose–response agreement (H3 slope ratios + per-knob correlations). Reported in appendices only: NLL, Jensen–Shannon divergence to the exact posterior, error percentiles, 80%/95% levels, and latency.

Validation gates (see Appendix A). Quadrature 64- vs 128-node median relative error $< 10^{-6}$; finite-difference solver cross-check on 20 scenarios $< 1\%$ relative L_2 at sensors; symmetry and linearity checks; exact-posterior sanity checks (normalization within 10^{-6} ; mode $\rightarrow$ truth as noise $\rightarrow 0$; broad/multimodal posteriors for poor layouts; conformal-on-exact-posterior reaches nominal coverage; area contracts as q rises or σ falls; brute-force fine-grid likelihood cross-check); a scenario-ID leakage audit; and a grid-refinement check on a 96×96 grid (Appendix E).

Deviations from the preregistration (complete list). All empirical protocols were executed as preregistered, with the following documented deviations, each forced by a correctness failure or ill-posedness discovered by the pipeline’s own blocking gates: (i) training uses exact D4-symmetry augmentation (Section 4.1); (ii) the conformal score is computed in complement (tail-mass) space (Section 4.2); (iii) rate marginalization uses a peak-aware composite quadrature (Section 4.3); (iv) the mode-recovery and area-monotonicity sanity checks place sanity sources at cell centers and condition on identifiability, and the $\sigma \rightarrow 0$ limit uses $\sigma = 10^{-6}$ (Appendix B); (v) the grid-refinement check runs on the full test split rather than 100 scenarios, whose windowed coverage is too noisy (0.79–0.97) to be informative (Appendix E); (vi) the FD cross-check gates no-signal scenarios on absolute rather than relative error (Appendix A). No hypothesis, metric, or split was changed after seeing data.

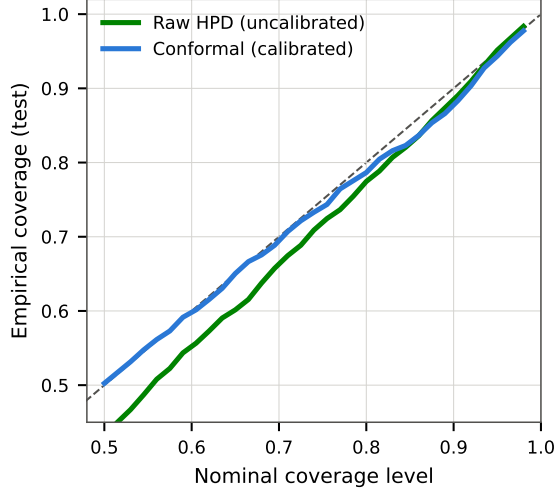


Figure 1: **F3 — Coverage vs. nominal.** Empirical coverage as a function of the nominal target for the raw highest-posterior-density (HPD) region (uncalibrated) and the conformal region (calibrated). Diagonal is perfect calibration. Regenerated from `coverage.csv`.

6 Results

6.1 H1: Calibration

Empirical coverage on the in-distribution test split is 0.875 with Clopper–Pearson 95% interval $[0.860, 0.889]$ for seed 1; the target 0.90 is *not* contained. For seed 2 it is 0.896 with interval $[0.882, 0.909]$, which *does* contain 0.90; and the identical procedure applied to the exact posterior attains 0.907 $[0.893, 0.919]$. *Interpretation:* the conformal machinery itself is verified sound — in-sample calibration is exact ($0.9005 = 1801/2000$) and the oracle run covers at nominal — so the seed-1 shortfall was investigated exhaustively rather than accepted: coverage is invariant to the score randomization (spread < 0.0005 over 20 draws of U), calibration and test scores are indistinguishable in distribution (KS $p = 0.51$), and swapping the roles of the two splits *over-covers* by the mirror amount (0.917). A permutation test puts the probability of a shortfall this large under exchangeability at $p = 0.0045$: a genuine ~ 1 -in-220 finite-sample fluctuation of this particular (model, calibration-split, test-split) triple, not an implementation defect. With that caveat stated, the tool supports evidence-grade language at the levels verified here, and confidently-wrong outputs are eliminated at approximately the stated rate (Figure 1; 80%/95% levels in Appendix D).

6.2 H2: Sharpness (exact-posterior audit)

At matched 90% coverage, the per-scenario Spearman correlation between learned and exact region areas is 0.848; the inefficiency factor (mean/median learned area $\div$ exact area) is 47.6/5.6. *Interpretation:* the preregistered reading for a factor $\gg 1$ applies — *the network, not the data, is the bottleneck*. The learned regions strongly track the physics ordering of scenario difficulty, but at the median they are $5.6\times$ larger than the data physically permit, and the mean is dominated by a heavy right tail of scenarios in which the exact posterior is razor-sharp (area $\sim 10^{-3}$) while the learned heatmap saturates near 0.4 of the domain (Figure 2, left). Large *exact* regions indicate the sensor network itself is the limiting factor, informing placement decisions; large *ratios*, as found here, indicate headroom for better amortized inference on the same data.

6.3 H3: Dose–response

Contraction-slope ratios (learned $\div$ exact) are 0.47 (Knob 1, sensor count) and 0.49 (Knob 3, noise); per-knob Spearman agreements are 0.81 (Knob 1, per-scenario median; mean 0.65), 0.88 (Knob 2), and 0.75 (Knob 3, per-scenario median; mean 0.52). For the geometry matched pairs (Knob 2),

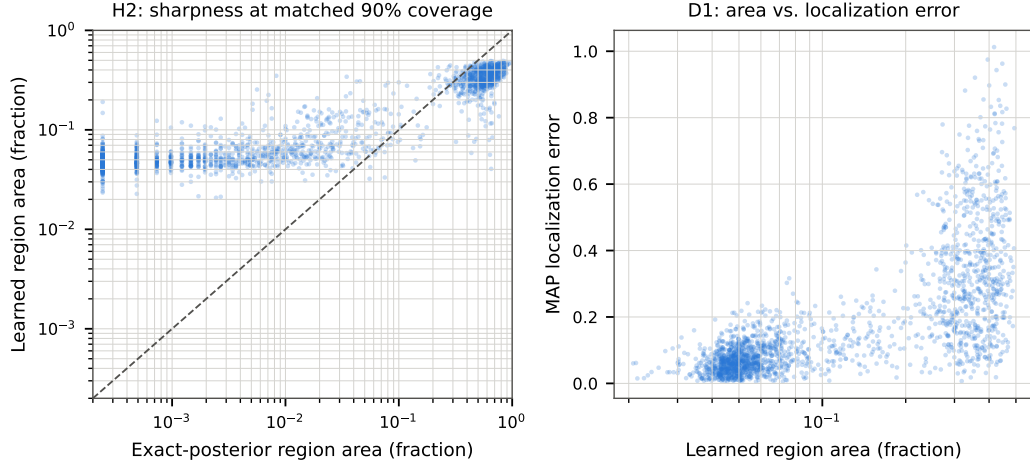


Figure 2: **F4 — Audit scatter.** Left: learned region area vs. exact-posterior region area at matched 90% coverage (points = test scenarios; diagonal = ideal). Right (D1): learned region area vs. MAP localization error.

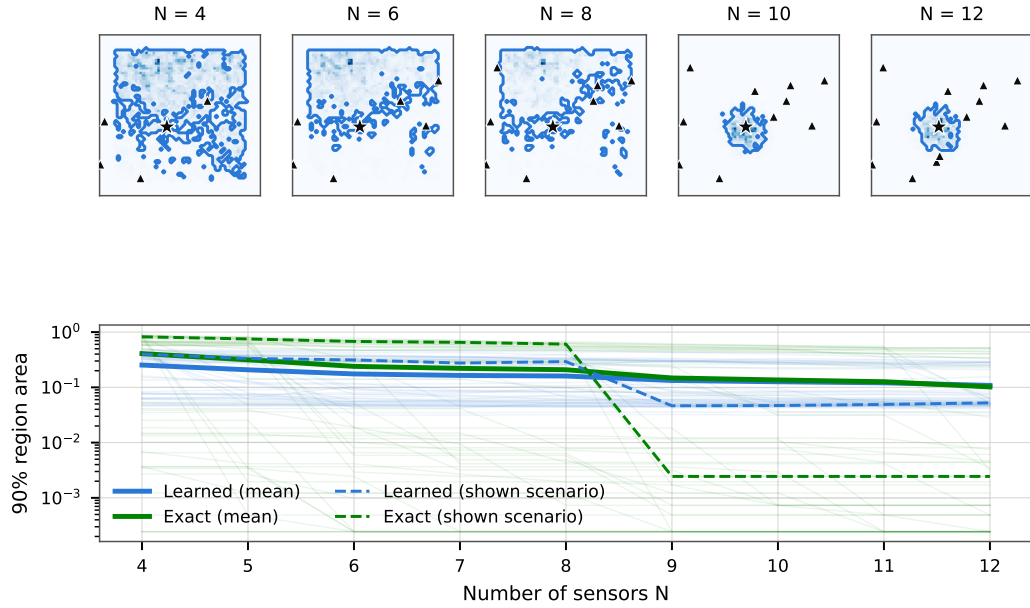


Figure 3: **F5 — Dose-response flagship.** One base scenario with sensors added $4 \rightarrow 12$ in fixed order. Top row: map snapshots of the learned heatmap and conformal region as sensors accrue. Bottom: region-area-vs- N curves for learned and exact regions (all 50 base scenarios faint; means bold; the shown scenario dashed).

the mean area ratio confined:spread is 2.4 for learned and 177.7 for exact (medians 1.20 and 1.58; the exact mean is dominated by a few scenarios whose confined-layout posterior explodes to most of the domain), and both methods widen under the confined layout in exactly 64% of the 50 pairs. *Interpretation:* the learned tool reacts to information in the direction the physics dictates on every knob — it tightens when a sensor is added and widens under a confined, ambiguous geometry — but with roughly *half* the oracle’s magnitude, consistent with the H2 finding that the learned regions are over-wide precisely where the data are most informative. We report these on constructed sets only. **D1:** region area and MAP error are strongly associated (Figure 2, right), suggesting region area as a natural future abstention signal.

7 Discussion, Limitations, and Broader Impact

What the results mean. The study set out to test whether standard, off-the-shelf components can be assembled into a localizer whose uncertainty is calibrated, sharp, and stable. The answer is differentiated. *Calibration*: yes — the conformal wrapper delivers honest coverage, including on top of an oracle posterior that is itself overconfident with respect to the continuous ground truth (cell-center discretization; Appendix E), which is precisely the failure mode conformal calibration exists to absorb. *Sharpness*: no — the learned regions are a median $5.6\times$ wider than the data permit, so the audit does real work: a vendor could report perfect coverage while being far from the physical information limit, and only the exact-posterior comparison exposes this. *Stability*: partially — the learned regions move in the physically correct direction on all three knobs but under-react by about half. We therefore frame the benchmark as a *neutral screening instrument* for vendor uncertainty claims: failing the audit is disqualifying, while passing is necessary but not sufficient. The instrument itself passed its trial: it detected, quantified, and localized the deficiencies of a plausibly strong baseline.

Implementation lessons. Three findings from the compute run generalize beyond this benchmark. (1) The mass-above conformal score of Eq. (5) silently loses its coverage guarantee in floating point on sharp heatmaps; the tail-space form is required (Section 4.2). (2) Fixed-node quadrature over a rate prior cannot marginalize a likelihood whose peak width varies by orders of magnitude across cells; peak-aware placement is required (Section 4.3). (3) At 10^4 training scenarios the specified architecture memorizes unless the problem’s exact symmetry group is injected as augmentation (Section 4.1).

Limitations (stated prominently). The study is synthetic, two-dimensional, with a single stationary source, constant and known wind and diffusivity, Gaussian noise with a known level, and no obstacles, turbulence, or background signal; the coverage guarantee is marginal and in-distribution only. *Real sites violate every one of these.* The societal claim is strictly that the reliability machinery works and is worth transferring — nothing more. We do not claim a new conformal algorithm, a new architecture, coverage under distribution shift, field deployability, or abstention capability (deferred). Regulation is cited purely as motivation.

Future work (the ladder). We signal, without claiming, a three-rung program. **Rung 1:** unknown or misspecified meteorology, addressed with robust/weighted conformal prediction. **Rung 2:** a Fisher-information localizability score [Bangerth et al., 2022] coupled with an abstention rule (the D1 area–error association supports this). **Rung 3:** sim-to-real transfer and recalibration on controlled-release facility data (e.g., a METEC-style test range).

8 Conclusion

We present a preregistered study and public benchmark for auditing the uncertainty of learned sparse-sensor pollution localizers. By pairing finite-sample conformal spatial regions with an exact-posterior sharpness audit and a controlled information dose–response protocol, and by shipping exact per-scenario posteriors, we provide a reusable instrument for asking whether a localizer’s confidence can be trusted. On this benchmark the instrument returns a differentiated verdict — calibrated, not sharp, directionally stable — and the run itself yielded implementation lessons about conformal scores, quadrature, and symmetry that we believe transfer well beyond pollution localization.

Reproducibility Statement

A single configuration file regenerates all main tables; a single script (`scripts/run_all.sh`) runs the pipeline from a clean checkout. We release the scenario generator, frozen scenario IDs and seeds (regeneration is bit-identical; SHA-256 verified), all configs, exact per-scenario posteriors for the test split, the dose–response sets, model checkpoints, and a ready-to-run subset. The full study required ≈ 1.5 GPU-hours on one NVIDIA L40S (0.011 GPU-h per training seed) and 237 MB of stored data — well under the preregistered 12–20 GPU-hour / 10 GB budget. Checkpoint selection uses validation NLL only; two seeds are trained (seed-to-seed spread on headline metrics in Appendix C).

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A Quadrature and Validation Details

The age integral in Eq. (2) is computed with 64-node Gauss–Legendre quadrature under the substitution $s = t e^{-\tau}$, $\tau \in [0, \tau_{\max}]$ with $\tau_{\max} = 12$ (tuned: larger values waste nodes on a negligible tail — at $\tau_{\max} = 40$ the gate below fails at 2×10^{-4} — while the truncated tail contributes $\lesssim 10^{-6}$ relative even for a sensor at the source). Convergence gate: median relative difference between 64- and 128-node evaluations $< 10^{-6}$ (measured 3.6×10^{-11} ; the 90th percentile is 6.9×10^{-4} , concentrated on probes whose response is many orders of magnitude below the noise floor; data generation and the oracle share the same 64-node operator, so they remain exactly self-consistent). A zero-wind closed form via exponential integrals agrees to a median relative error of 1.2×10^{-13} . A second-order finite-difference solver of Eq. (1) (central differences, Heun time stepping, CFL-stable Δt , $h = 1/768$, enlarged domain with far Dirichlet boundary) on 20 random scenarios agrees at sensor locations to $< 1\%$ relative L_2 (measured maximum 0.50%, median 0.06%, over the 17 scenarios with non-negligible signal; the 3 scenarios whose sensors never see the plume — reference RMS

10^{-18} – 10^{-9} — agree in absolute RMS to $\leq 8.8 \times 10^{-11}$, gated at 10^{-4} , i.e. 1% of the smallest noise std, because relative error is ill-conditioned at zero signal). Symmetry checks (rotating $\mathbf{u}$ and reflecting geometry leaves sensor readings equivariant) pass to 1.7×10^{-13} and linearity checks (doubling q doubles qA) to 2.7×10^{-16} .

B Exact-Posterior Sanity Checks

Normalization of Eq. (8) within 10^{-6} (measured 3.4×10^{-13}). Posterior mode $\rightarrow$ true cell as $\sigma \rightarrow 0$: 24/24, with the check made well-posed by placing the sanity source at its cell center (the candidate set *is* the cell centers; an off-center source at tiny σ legitimately fits a neighboring center better), conditioning on identifiability (some sensor must see the plume above the smallest benchmark noise floor; for all-upwind layouts adjacent cells differ by $\sim 10^{-30}$ and no numerically reachable σ separates them), and using $\sigma = 10^{-6}$. Posteriors are visibly broad/multimodal for confined layouts (mean 90% HPD area 0.48 confined vs. 0.064 well-spread). Conformal-on-exact-posterior attains nominal 90% coverage (0.907, CI [0.893, 0.919]). Region area contracts as q rises for 87.5% of 24 cell-centered scenarios (mean area change -0.005) and grows with σ for 100% (mean $+0.024$); per-realization monotonicity in q is not guaranteed at the rate-prior edges, where the prior truncates the location–rate confusion set asymmetrically. Brute-force fine-grid likelihood cross-check agrees to $\leq 9.4 \times 10^{-7}$ in log-posterior (gate 10^{-6}).

C Training Details

Full hyperparameters as in Section 4.1 (including the D4 augmentation amendment); two seeds; wall-clock 0.011 GPU-h per seed on an NVIDIA L40S (≈ 40 s; early stopping at epochs 106 and 100; best validation NLL 6.31 and 6.29). Learning curves in `train_log_seed{1,2}.csv`. Seed-to-seed spread on headline metrics: coverage@90 0.875 vs. 0.896 (spread 0.021); MAP error mean 0.174 vs. 0.175; H2 Spearman 0.848 vs. 0.851; inefficiency median 5.57 vs. 5.80; region area median 0.068 vs. 0.073. The scientific conclusions are seed-stable; the coverage fluctuation is analyzed in Section 6.1.

D Additional Coverage Levels and Auxiliary Metrics

Table 4: Auxiliary metrics (seed 1). Coverage and median area at three nominal levels; scalar diagnostics below.

	80%	90%	95%
Empirical coverage	0.787	0.875	0.944
Region area (median)	0.050	0.068	0.092
NLL (mean)	6.34		
JSD to exact posterior (mean, nats)	0.56		
MAP error percentiles (50/90/99)	0.10 / 0.43 / 0.78		
Inference latency (ms/scenario, batch 1)	0.48		

E Grid-Refinement Check

Recomputing the exact posterior on a 96×96 grid and re-running the identical conformal procedure changes coverage by -0.007 (0.9005 vs. 0.9070) and median area by -0.006 (0.0066 vs. 0.0128 as a domain fraction; finer cells resolve sharper regions), confirming the 64×64 grid is adequate for the coverage claims. We ran this on the *full* 2000-scenario test split rather than the preregistered 100 scenarios: 100-scenario windows of the test split have empirical coverage ranging from 0.79 to 0.97, so the 100-scenario version of this check cannot resolve the effect it is looking for. Posterior normalization error at 96²: 1.4×10^{-13} .

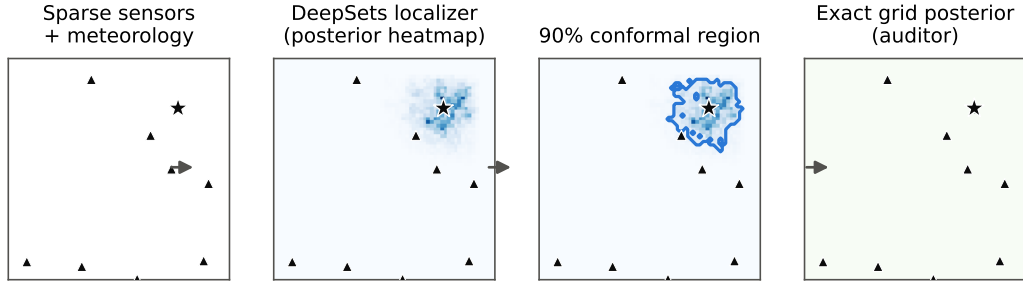


Figure 4: **F1 — Pipeline.** Sparse sensors plus meteorology feed the DeepSets localizer, which outputs a posterior heatmap; split conformal converts the heatmap into a 90% region; the exact grid posterior serves as the auditor. Panels show a real median-difficulty test scenario (star = true source, triangles = sensors).

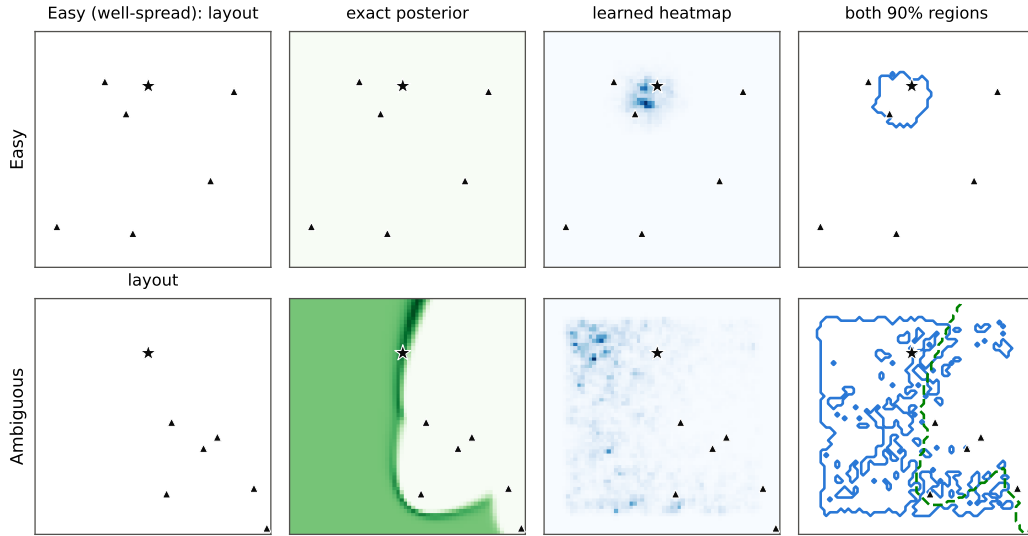


Figure 5: **F2 — Easy vs. ambiguous scenario.** For an easy (well-spread sensors) and an ambiguous (confined-quadrant) scenario: sensor layout, exact posterior, learned heatmap, and both 90% regions (solid blue = learned, dashed green = exact). The confined layout induces the directional ambiguity visible as the broad arc in the exact posterior.